

HTML - Comments

Comment is a piece of code which is ignored by any web browser. It is a good practice to add comments into HTML code, especially in complex documents, to indicate sections of a document, and any other notes to anyone looking at the code. Comments help to understand code and increases code readability.

HTML comments are placed in between `<!-- ... -->` tags. So, any content placed with-in `<!-- ..... -->` tags will be treated as comment and will be completely ignored by the browser.

Example

```
<!DOCTYPE html>
<html>

<head> <!-- Document Header Starts -->
  <title>This is document title</title>
</head> <!-- Document Header Ends -->

<body>
  <!--
    This is a multiline comment and it can
    span through as many as lines you like.
  -->
  <p>Document content goes here.....</p>
</body>

</html>
```

Valid vs Invalid Comments

Comments do not nest which means a comment cannot be put inside another comment.

Second the double-dash sequence `--` may not appear inside a comment except as part of the closing `-->` tag and there are no spaces in the start-of comment string.

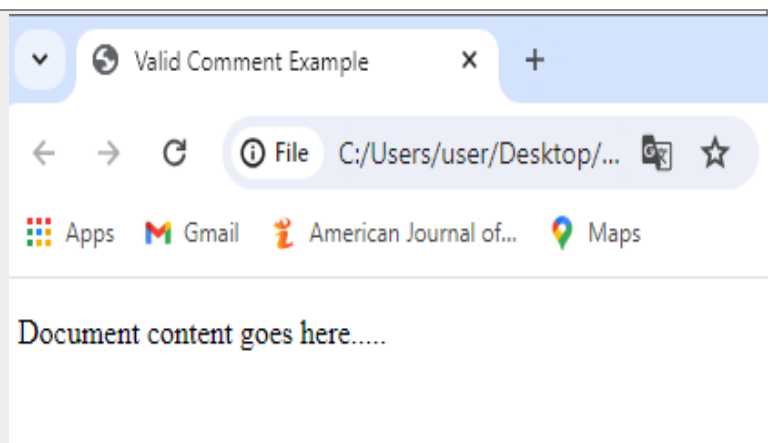
Example

Here, the given comment is a valid comment and will be wiped off by the browser.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Valid Comment Example</title>
  </head>

  <body>
    <!-- This is valid comment -->
    <p>Document content goes here.....</p>
  </body>

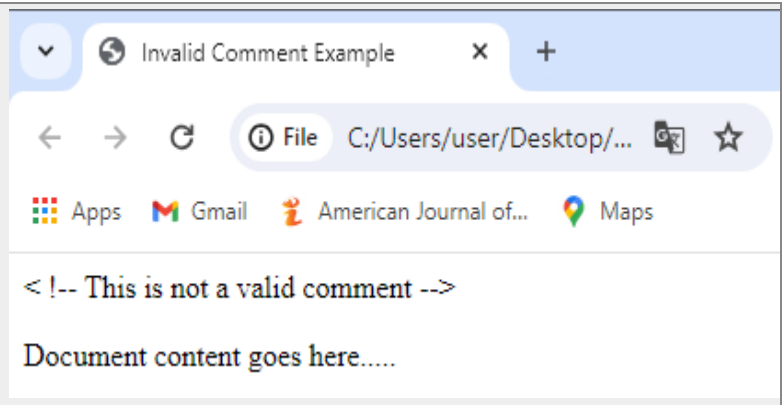
</html>
```



But, following line is not a valid comment and will be displayed by the browser. This is because there is a space between the left angle bracket and the exclamation mark.

```
<!DOCTYPE html>
<html>
  <head>
    <title>Invalid Comment Example</title>
  </head>

  <body>
    <!-- This is not a valid comment -->
    <p>Document content goes here.....</p>
  </body>
</html>
```



Multiline Comments

HTML supports multi-line comments as well. We can comment multiple lines by the special beginning tag `<!--` and ending tag `-->` placed before the first line and end of the last line as shown in the given example below.

Example

```
<!DOCTYPE html>
<html>

  <head>
    <title>Multiline Comments</title>
  </head>

  <body>
    <!--
      This is a multiline comment and it can
      span through as many as lines you like.
    -->

    <p>Document content goes here.....</p>
  </body>
</html>
```

Tag Attributes:

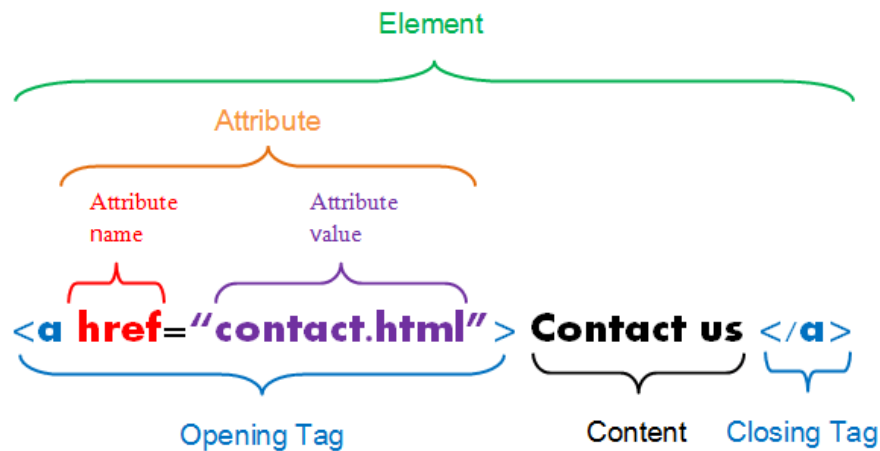
Tags can have attributes. The Attributes can provide additional information about the HTML elements on the page. The `<tag>` tells the browser to do something, while the attribute tells the browser how to do it. For instance, if we add the `bgcolor` attribute, we can tell the browser that the background color of the page should be blue, like this:

```
<body bgcolor="blue">.
```

This tag defines an HTML table: `<table>`. With an added `border` attribute, tell the browser that the table should have no borders:

<table border="0">.

Attributes always come in **name/value** pairs like this: **name="value"**. Attributes are always added to the start tag of an HTML element and the value is surrounded by quotes.



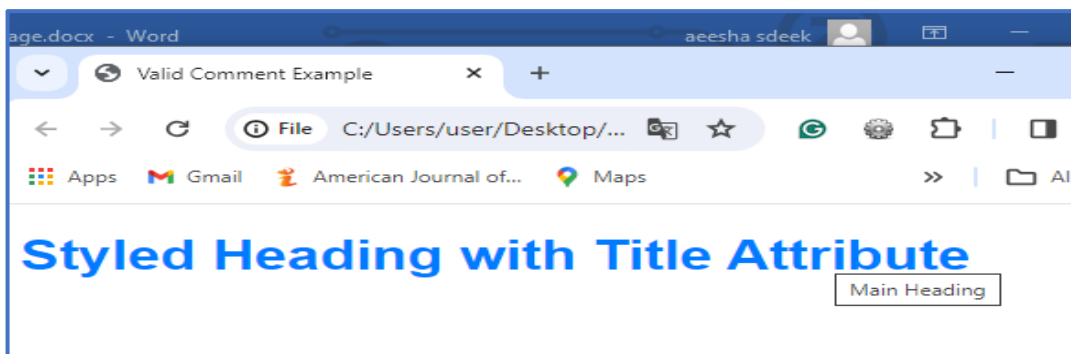
Quote Styles, "red" or 'red'?

Attribute values should always be enclosed in quotes. Double style quotes are the most common, but single style quotes are also allowed. In some rare situations, like when the attribute value itself contains quotes, it is necessary to use single quotes:

name='George "machine Gun" Kelly'

Example about attribute with header tag `<h1>` tag with the **style**, **title**, and **border** attributes:

`<h1 style="color: #007bff; " title="Main Heading" > Styled Heading with Title Attribute</h1>`

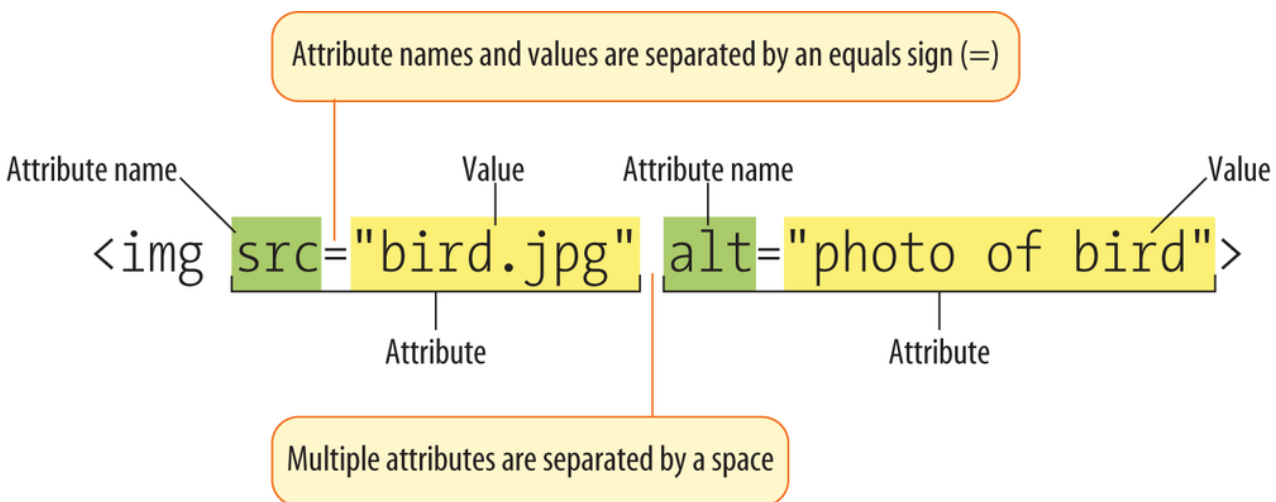


HTML - Images

Images are very important to beautify as well as to depict many complex concepts in simple way on web page.

Insert Image

We can insert any image in web page by using **** tag. Following is the simple syntax to use this tag.



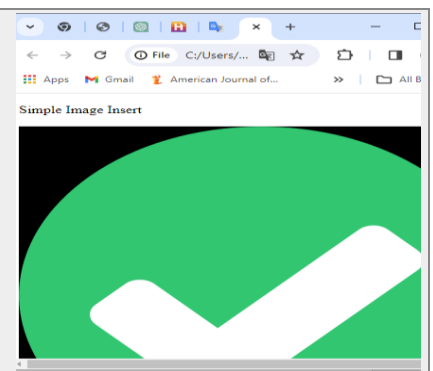
```
<img src = "Image URL" ... attributes-list/>
```

The **** tag is an empty tag, which means that, it can contain only list of attributes and it has no closing tag.

Example

keep HTML file test.htm and image file test.png in the same directory –

```
<!DOCTYPE html>
<html>
  <head>
    <title>Using Image in Webpage</title>
  </head>
  <body>
    <p>Simple Image Insert</p>
    <img src = "html/images/test.png" alt = "Test Image" />
  </body>
</html>
```



Images can be PNG, JPEG or GIF, put image file name in **src** attribute. Image name is always **case sensitive**.

The **alt** attribute is a mandatory attribute which specifies an alternate text for an image, if the image cannot be displayed.

Set Image Location

Usually keep all the images in a separate directory. keep HTML file test.htm in home directory and create a subdirectory image inside the home directory where will keep the image test.png.

Example

Assuming image location is "image/test.png", try the following example –

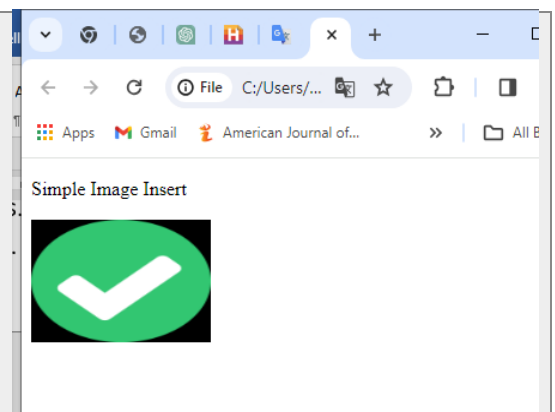
```
<!DOCTYPE html>
<html>
  <head>
    <title>Using Image in Webpage</title>
  </head>
  <body>
    <p>Simple Image Insert</p>
    <img src = "html/images/test.png" alt = "Test Image" />
  </body>
</html>
```

Set Image Width/Height

To set image width and height using **width** and **height** attributes. specify width and height of the image in terms of either pixels or percentage of its actual size.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Set Image Width and Height</title>
  </head>
  <body>
    <p>Setting image width and height</p>
    <img src = "html/images/test.png"
alt = "Test Image" width = "150" height = "100" />
  </body>
</html>
```



Set Image Border

By default, image will have a border around it, and can specify border thickness in terms of pixels using border attribute. A thickness of 0 means, no border around the picture.

Example

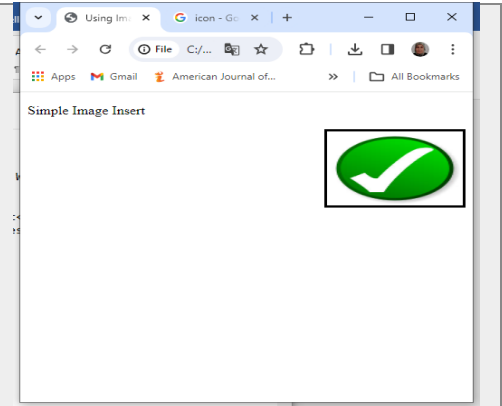
```
<!DOCTYPE html>
<html>
  <head>
    <title>Set Image Border</title>
  </head>
  <body>
    <p>Setting image Border</p>
    <img src = "html/images/test.png"
alt = "Test Image" border = "3" />
  </body>
</html>
```

Set Image Alignment

By default, image will align at the left side of the page, but can use **align** attribute to set it in the center or right.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>Set Image Alignment</title>
  </head>
  <body>
    <p>Setting image Alignment</p>
    <img src = "/html/images/test.png"
alt = "Test Image" border = "3" align = "right"/>
  </body>
</html>
```



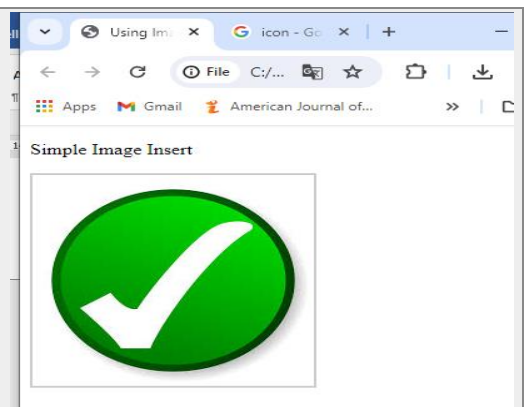
style with border: the attribute style can take different properties the first one is **border**

The **border** property is a shorthand property for (in the following order):

- border **width** take value:
- border **style** (required)
 1. **dotted** - Defines a dotted border
 2. **dashed** - Defines a dashed border
 3. **solid** - Defines a solid border
 4. **double** - Defines a double border
- border **color**

Example:

```
<!DOCTYPE html>
<html>
  <head>
    <title>Set Image style</title>
  </head>
  <body>
    <p>Setting image style</p>
    
  </body>
</html>
```



Explanation:

- **border:** Sets the border properties.
- **2px:** Specifies the border width (2 pixels in this example).
- **solid:** Sets the border style to a solid line.
- **#ccc:** Sets the border color to a light gray color. this can be replaced this with any valid CSS color value.

The **style** attribute allows to apply inline CSS directly to the **** tag.

We can apply specific styles to each side of the border using individual properties like:

➤ **border-left, border-right, border-top, and border-bottom**

Here's an example applying styles to the left side:

```

```

Explanation:

- **border-left:** Sets the border properties for the left side.
- **4px:** Specifies the border width for the left side (4 pixels in this example).
- **solid:** Sets the border style to a solid line.
- **#3498db:** Sets the border color to a specific color (can replace it with any valid CSS color value).

Also can apply similar styles to other sides by using:

border-right, border-top, and border-bottom.

Here's an example applying styles to all sides:

```

```

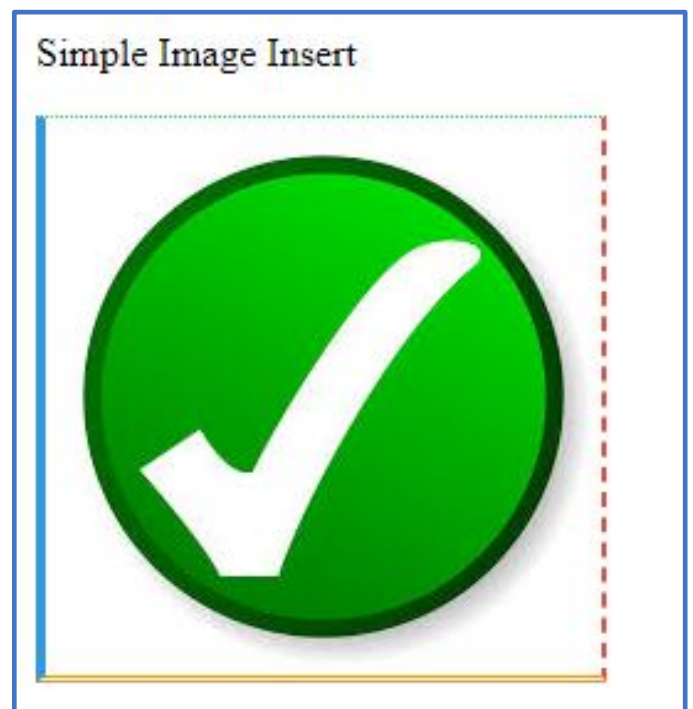
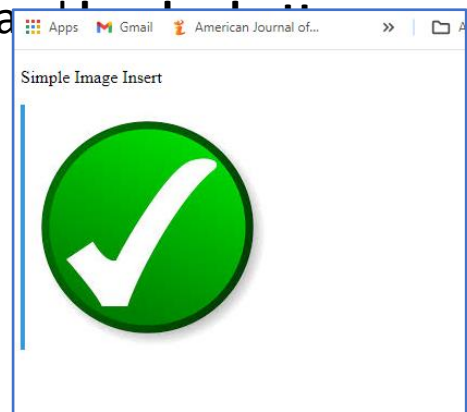
In this example:

- The left side has a 4-pixel solid border of color **#3498db**.
- The right side has a 2-pixel dashed border of color **#e74c3c**.
- The top side has a 1-pixel dotted border of color **#2ecc71**.
- The bottom side has a 3-pixel double border of color **#f39c12**.

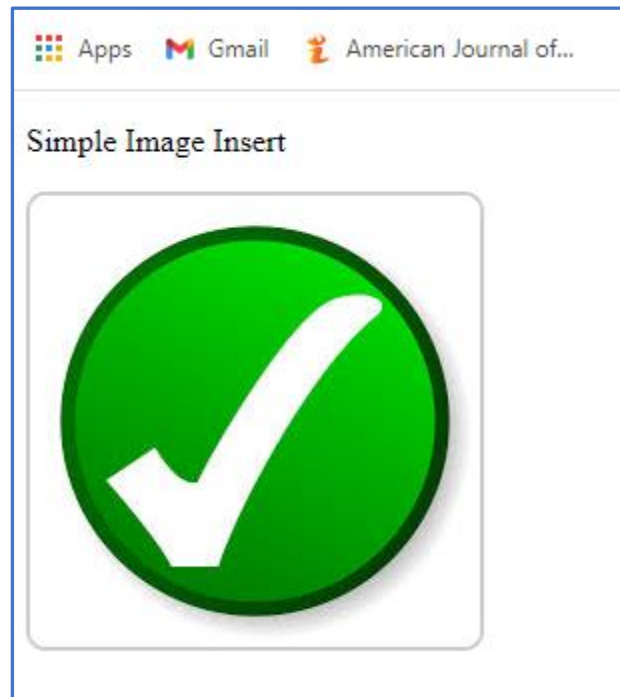
Other properties:

➤ **border-radius property:**

The **border-radius** property in CSS is used to create rounded corners for an element, including images. we can apply it to an image within the **style** attribute. Here's an example:



``

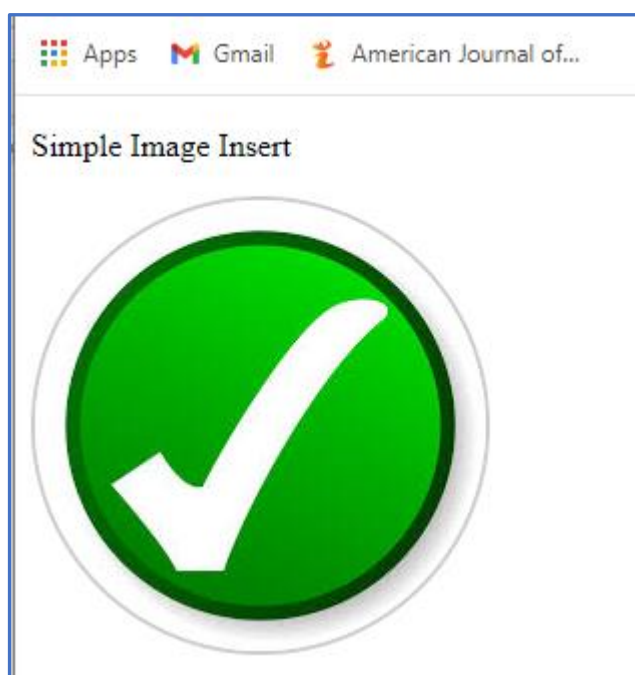


Explanation:

- **border-radius:** Sets the radius of the element's corners.
- **10px:** Specifies the radius size (10 pixels in this example).

This example adds a 10-pixel border-radius to all corners of the image, creating rounded corners. also can adjust the **border-radius** value to make the corners more or less rounded, and can combine it with other border styles and properties as needed.

`<img src = "html/images/im.jpeg" alt="Description" style="border: 2px solid #ccc; border-radius: 50%;" />`

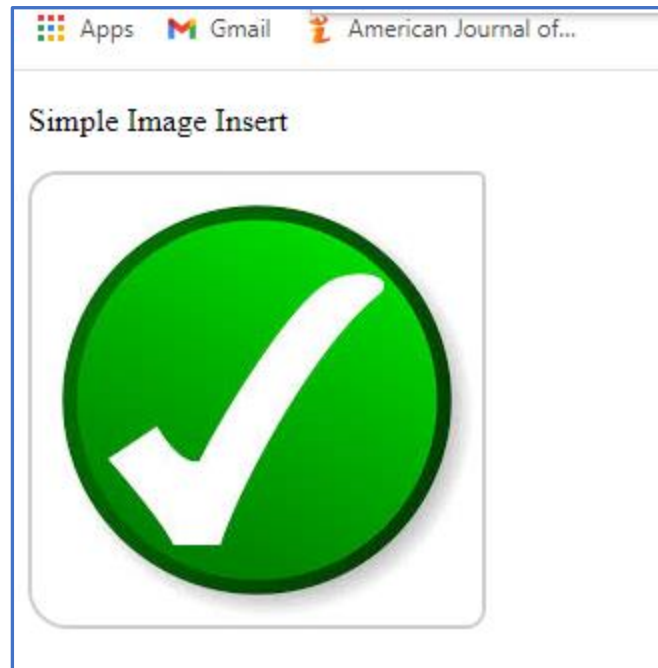


If we want to apply different radius values to each corner, we can use the **border-radius** property with four values representing the radius of the **top-left, top-right, bottom-right, and bottom-left corners**, respectively. For example:

```

```

In this example, the top-left corner has a radius of 15 pixels, the top-right has a radius of 5 pixels, the bottom-right has a radius of 10 pixels, and the bottom-left has a radius of 20 pixels.

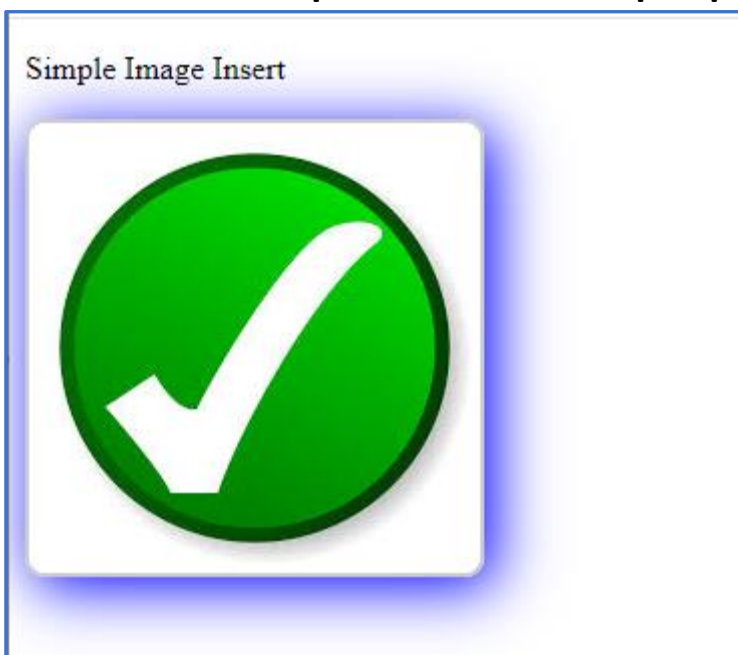


➤ **box-shadow property:**

the property called **box-shadow** in CSS. Here's an example:

```

```



Explanation:

- **box-shadow:** Adds a shadow effect to the element.
- **5px 5px:** The horizontal and vertical offset of the shadow.
- **10px:** The blur radius of the shadow.
- **rgba(0, 0, 0, 0.3):** The color and transparency of the shadow. In this example, it's a semi-transparent black shadow.

HTML - Marquees

An HTML marquee is a scrolling piece of text displayed either horizontally across or vertically down webpage depending on the settings. This is created by using HTML <marquees> tag.

Note – The <marquee> tag deprecated in HTML5. Do not use this element, instead can use JavaScript and CSS to create such effects.

Syntax

A simple syntax to use HTML <marquee> tag is as follows –

```
<marquee attribute_name = "attribute_value" ....more attributes>  
  One or more lines or text message or image  
</marquee>
```

The <marquee> Tag Attributes

Following is the list of important attributes which can be used with <marquee> tag.

Sr.No	Attribute & Description
1	width This specifies the width of the marquee. This can be a value like 10 or 20% etc.
2	height This specifies the height of the marquee. This can be a value like 10 or 20% etc.
3	direction This specifies the direction in which marquee should scroll. This can be a value like <i>up</i> , <i>down</i> , <i>left</i> or <i>right</i> .
4	behavior

	This specifies the type of scrolling of the marquee. This can have a value like <i>scroll</i> , <i>slide</i> and <i>alternate</i> .
5	loop This specifies how many times to loop. The default value is INFINITE, which means that the marquee loops endlessly.
6	bgcolor This specifies background color in terms of color name or color hex value.

Examples - 1

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML marquee Tag</title>
  </head>
  <body>
    <marquee>This is basic example of marquee</marquee>
    <marquee width = "50%">This example will take only 50% width</marquee>
  </body>
</html>
```

Examples - 2

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML marquee Tag</title>
  </head>
  <body>
    <marquee direction = "right">This text will scroll from left to right</marquee>
    <marquee behavior="scroll" direction="up" >Slow Scrolling</marquee>
    <marquee behavior="scroll" direction="right" >Little Fast Scrolling</marquee>
    <marquee behavior="scroll" direction="left" >Fast Scrolling</marquee>
    <marquee behavior="scroll" direction="right" >Very Fast Scrolling</marquee>
  </body>
</html>
```

Examples - 4

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML marquee Tag</title>
  </head>
  <body>
    <marquee height=25px bgcolor="Green" direction="left" >Left </marquee>
    <marquee height=50px bgcolor="Green" direction="right" >Right </marquee>
    <marquee direction = "up">This text will scroll from bottom to up</marquee>
  </body>
</html>
```